chAInge

Project Overview

Team: Jet2Holiday

1. Problem and Scenario

Enterprise upgrades frequently trigger shifting policies, product specifications, and operational processes. Updating the vast array of corresponding brochures, websites, and internal handbooks typically requires complex, cross-departmental coordination that can drain weeks of manual effort. This tedious process is highly vulnerable to human error, resulting in critical data inconsistencies that expose businesses to customer disputes and severe compliance risks.

2. Core Solution

To tackle this, we developed a multi-agent system designed to auto-search, update, and verify, thus compressing update cycles from several days down to mere minutes. Our architecture leverages open-source models deployed locally, eliminating the need for internet connectivity. This design choice architecturally ensures that corporate data never leaks outside the secure environment.

Every file ingested into the RAG database is tagged with metadata-based permission levels, while a dedicated Permission Audit Agent evaluates each Worker response based solely on the user's role and the generated output. Critically, human conversational history is never fed into this audit, effectively neutralizing injection attacks.

Unlike conventional RAG systems that merely return top K results, our solution prioritizes high recall and deep cross-file impact analysis. It supports Multi-Search, Exact Search, and Full Document Retrieval, enabling agents to aggregate findings from multiple angles, deduplicate results, and thoroughly read entire documents to eliminate oversight risks.

To orchestrate this complex flow, we designed an Enterprise-RAG Skill that guides agents through multi-round retrieval and rigorous consistency checks. It clearly distinguishes verifiable corporate facts from AI-generated reasoning, seamlessly fusing the knowledge base, RAG retrieval, specialized Skills, and multi-agent collaboration into a cohesive, end-to-end workflow.

3. Innovation and Differentiating Advantages

The Verification Agent provides a "safety-by-design" validation layer, ensuring zero critical failures breach production:

- Automated Guardrails: Scans assets autonomously to block data anomalies and unauthorized document access.
- Proactive Conflict Detection: Automatically flags margin breaches, data inconsistencies, and downstream dependency errors.

- Human-in-the-Loop Oversight: Mandates human approvals for high-risk actions, blending automated speed with expert validation.

4. Open-Source / Reusability Value

The platform transforms raw domain expertise into deterministic, open-source-ready AI skills through a highly decoupled, auditable multi-agent framework:

- Cross-Border Scaling: Automates workflows across global entities while dynamically adapting to regional regulations.
- Skill Institutionalization: Converts complex corporate expertise into reusable AI capabilities using structured verification frameworks.
- Precise Orchestration: Leverages semantic routing, robust tool-calling, and fully traceable audit logs.
- Modular Architecture: Separates the core pipeline into independent Python modules to isolate code, data, and evaluation.
- Configuration Decoupling: Isolates sensitive application logic by storing credentials and Qdrant database metadata in separate directories.
- Community Readiness: Standardizes agent skill schemas, Model Context Protocol (MCP) configs, and ingestion workflows for open-source adoption.

5. Current Progress

Phase 1 is complete, establishing a secure data foundation and a functional multi-agent baseline:

- Data Scale: Ingested 15,000 corporate documents, processing them into 44,000 cleaned, vectorized chunks securely stored in Qdrant.
- Cloud Architecture: Engineered a custom cloud RAG database optimized for high-volume enterprise knowledge.
- Secure Access: Deployed a lightweight multi-agent infrastructure supporting secure user login and interaction via SSH.
- Standardized Assets: Developed foundational agent skill schemas and validated a dynamic multi-agent file-lookup demo.